Tugas 17/03/26

1. $\int e^{2x} \underbrace{\sin 3x}_{u} \, dx$

$u = \sin 3x$ $\qquad$ $dv = e^{2x} dx$

$du = 3 \cos 3x \, dx$ $\qquad$ $v = \frac{1}{2} e^{2x} + C$

$\int e^{2x} \sin 3x \, dx = \frac{1}{2} e^{2x} \sin 3x - \int \frac{1}{2} e^{2x} \cdot 3 \cos 3x \, dx$

$\int e^{2x} \sin 3x \, dx = \frac{1}{2} e^{2x} \sin 3x - \frac{3}{2} \int e^{2x} \cos 3x \, dx$

$$\int e^{2x} \underbrace{\cos 3x}_{u} \, dx$$

$$u = \cos 3x \qquad dv = e^{2x}\,dx$$
$$du = -3 \sin 3x \, dx \qquad v = \frac{1}{2} e^{2x} + c$$

$$= \frac{1}{2} e^{2x} \cdot \cos 3x - \int \frac{-3}{2} e^{2x} \sin 3x \, dx$$
$$= \frac{1}{2} e^{2x} \cos 3x + \frac{3}{2} \int e^{2x} \sin 3x \, dx$$

$$\int e^{2x} \sin 3x \, dx = \frac{1}{2} e^{2x} \sin 3x - \frac{3}{2} \left[\frac{1}{2} e^{2x} \cos 3x + \frac{3}{2} \int e^{2x} \sin 3x \, dx \right]$$

$$\int e^{2x} \sin 3x \, dx = \frac{1}{2} e^{2x} \sin 3x - \frac{3}{4} e^{2x} \cos 3x - \frac{9}{4} \int e^{2x} \sin 3x \, dx$$

$$\frac{13}{4} \int e^{2x} \sin 3x \, dx = \frac{1}{2} e^{2x} \sin 3x - \frac{3}{4} e^{2x} \cos 3x$$

$$\int e^{2x} \sin 3x \, dx = \frac{4\left[2 \cdot \frac{1}{2} e^{2x} \sin 3x - \frac{3}{4} e^{2x} \cos 3x\right]}{13}$$

$$\int e^{2x} \sin 3x \, dx = \frac{2e^{2x} \sin 3x - 3 e^{2x} \cos 3x}{13}$$

2. $\int x \cos^2(x) \, dx$

$$\int x \left(\frac{1 + \cos 2x}{2} \right) dx$$
$$\frac{1}{2} \int x \left(1 + \cos 2x \right) dx$$
$$\frac{1}{2} \int x + \frac{1}{2} \int x \cos 2x \, dx$$
$$\frac{x^2}{4} + \frac{1}{2} \int \underbrace{x}_{u} \underbrace{\cos 2x \, dx}_{dv}$$
$$du = dx \qquad v = \frac{1}{2} \sin 2x$$
$$\frac{x^2}{4} + \frac{1}{2} \left(\frac{x}{2} \sin 2x - \frac{1}{2} \int \sin 2x \, dx \right)$$
$$\frac{x^2}{4} + \frac{1}{2} \left(\frac{x}{2} \sin 2x - \frac{1}{2} \left(-\frac{1}{2} \cos 2x \right) \right)$$
$$\frac{x^2}{4} + \frac{1}{2} \left(\frac{x}{2} \sin 2x + \frac{1}{4} \cos 2x \right)$$
$$\frac{x^2}{4} + \frac{1}{4} x \sin 2x + \frac{1}{8} \cos 2x + C$$

3. $\displaystyle\int \frac{11-2x}{x^2+3x-10}\,dx$

$$\frac{11-2x}{(x+5)(x-2)} = \frac{A}{x+5} + \frac{B}{x-2}$$

$$\frac{11-2x}{(x+5)(x-2)} = \frac{A(x-2)+B(x+5)}{(x+5)(x-2)}$$

$11-2x = A(x-2)+B(x+5)$

$x=2$

$11-4 = B(2+5)$

$7 = 7B$

$B=1$

$x=-5$

$11-2(-5) = A(-5-2)$

$21 = -7A$

$A=-3$

$$\frac{11-2x}{(x+5)(x-2)} = \frac{-3}{x+5} + \frac{1}{x-2}$$

$$\int \frac{11-2x}{(x+5)(x-2)}\,dx = -3\int \frac{1}{x+5}\,dx + \int \frac{1}{x-2}\,dx$$

$$= -3\ln|x+5| + \ln|x-2| + C$$

4. $\displaystyle\int \frac{x-7}{6x^2+x-1}\,dx$

$$\frac{x-7}{6x^2+x-1} = \frac{x-7}{(2x+1)(3x-1)} = \frac{A}{2x+1} + \frac{B}{3x-1}$$

$$\frac{x-7}{(2x+1)(3x-1)} = \frac{A(3x-1)+B(2x+1)}{(2x+1)(3x-1)}$$

$x-7 = A(3x-1)+B(2x+1)$

$x=-\frac{1}{2}$

$-\frac{1}{2}-7 = A(-\frac{5}{2})$

$-\frac{15}{2} = -\frac{5}{2}A$

$-15 = -5A$

$A=3$

$x=1/3$

$1/3-7 = B(2/3+1)$

$-\frac{20}{3} = \frac{5}{3}B$

$-20 = 5B$

$B=-4$

$$\int \frac{x-7}{6x^2+x-1}\,dx = \int \frac{3}{2x+1} + \frac{-4}{3x-1}\,dx$$

$$= 3\int \frac{1}{2x+1}\,dx + (-4)\int \frac{1}{3x-1}\,dx$$

$$= 3\cdot\frac{\ln|2x+1|}{2} - 4\cdot\frac{\ln|3x-1|}{3}$$

$$= \frac{9\ln|2x+1| - 8\ln|3x-1|}{6} + C$$

5. $\displaystyle\int \frac{\cos x\,dx}{\sin^2 x - 6\sin x - 16}$

$$\frac{\cos x}{\sin^2 x - 6\sin x - 16} = \ldots$$

$a=\sin x$

$$\frac{\cos x}{a^2-6a-16} = \frac{\cos x}{(a-8)(a+2)}$$

$$= \frac{\cos x}{(-8+\sin x)(2+\sin x)} = \frac{A}{-8+\sin x} + \frac{B}{2+\sin x}$$

$$\frac{\cos x}{\sin^2 x - 6\sin x - 16} = \frac{A(2+\sin x)+B(-8+\sin x)}{(-8+\sin x)(2+\sin x)}$$

$$\int \frac{\cos x\,dx}{\sin^2 x - 6\sin x - 16} = \int \frac{A(2+\sin x)+B(-8+\sin x)}{(-8+\sin x)(2+\sin x)}\,dx$$

Misal $u=\sin x$

$du = \cos x\,dx$

$$\int \frac{du}{u^2-6u-16} = \int \frac{A(2+u)+B(-8+u)}{(-8+u)(2+u)}\,du$$

$1 = A(2+u)+B(-8+u)$

$u=-2$

$1 = B(-8+(-2))$

$B = -\frac{1}{10}$

$u=8$

$1 = A(10)$

$A = \frac{1}{10}$

$$\int \frac{du}{u^2-6u-16} = \int \frac{1/10}{-8+u}\,du + \int \frac{-1/10}{2+u}\,du$$

$$= \frac{1}{10}\left[\int \frac{du}{-8+u} - \int \frac{du}{2+u}\right]$$

$$= \frac{1}{10}\left[\ln|u-8| - \ln|u+2|\right]$$

$$= \frac{1}{10}\left[\ln|-8+\sin x| - \ln|2+\sin x|\right] + C$$